

# Design and analysis on Mach-Zehnder Interferometer

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**Abstract:** This report presents several implementations of Mach-Zehnder Interferometer based on Si based Strip waveguide with two different configurations (Y-splitter and broadband directional coupler). Transmission spectra of different MZIs are simulated for different optical path length difference between the two arms of the MZI to understand its impact on gain and free spectral range (FSR). An initial draft of the layout is presented for fabrication and automated measurement.

## 1. Introduction

Mach-Zehnder interferometer (MZI) is one of the most widely implemented interferometric architectures due to its spatially separated two-path geometry, high phase sensitivity, and design flexibility [1]. In an MZI, an incident coherent light beam is divided by an input splitter into two distinct arms and subsequently recombined via an output coupler. Any relative optical path delay or refractive index perturbation between the arms induces a differential phase shift ( $\Delta\phi$ ), yielding an interference pattern governed by a cosine-squared intensity distribution [2, 3]. With the rapid advancement of integrated photonics, MZIs fabricated on silicon-on-insulator (SOI) and silicon nitride (Si<sub>3</sub>N<sub>4</sub>) platforms have largely replaced bulk-optic arrangements, offering high mechanical stability and CMOS compatibility [4].

In this article, we present a few implementations of Mach-Zehnder Interferometer with two different configurations having Y-splitter and broadband directional coupler as the input splitter and output combiner. The objective is to understand the impact of one of the design parameters, namely the path length difference between the two arms on the transmission spectrum of MZI particularly FSR. The relevant mode simulation and characteristics of the waveguide used in MZI and draft layout of the designed MZIs are shown in the subsequent sections.

## 2. Theory

One of the most common implementations of Mach-Zehnder Interferometer is shown in Fig1 It consists of a split at the input into upper and lower waveguide portions and later combined at the output. The splitting and combining can be achieved by any types of splitters/combiners such as Y-branches

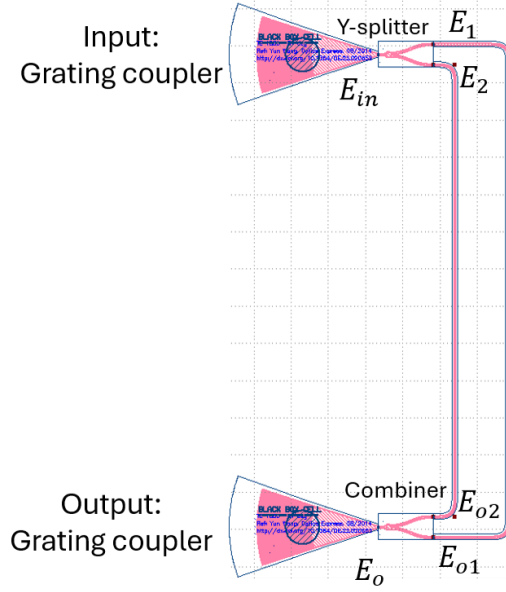


Fig1: Mach-Zehnder Interferometer implemented with grating coupler, Y-branch as splitter, waveguide and Y-branch as combiner.

As indicated in Fig1, the input intensity,  $I_{in}$  is split into half at the two output branches of the Y-splitter and thus the electric field at these two branches would be:

$$E_1 = \frac{E_{in}}{\sqrt{2}} \text{ and } E_2 = \frac{E_{in}}{\sqrt{2}}$$

Assuming identical waveguides (same effective index  $n_{eff}$  and propagation constant  $\beta = \frac{2\pi n_{eff}}{\lambda_o}$ ;  $\lambda_o$  is the free space wavelength) at left and right branches with length of  $L_1$  and  $L_2$  with negligible losses, the output electric field would be:

$$E_o = \frac{E_{o1}}{\sqrt{2}} + \frac{E_{o2}}{\sqrt{2}} = \frac{1}{\sqrt{2}} (E_1 e^{j\beta L_1} + E_2 e^{j\beta L_2}) = \frac{E_{in}}{2} (e^{j\beta L_1} + e^{j\beta L_2})$$

Thus the output intensity  $I_{out}$  would be:

$$I_{out} = \frac{I_{in}}{4} |e^{j\beta L_1} + e^{j\beta L_2}|^2 = \frac{I_{in}}{4} |e^{j\beta L_1} + e^{j\beta L_2}|^2 = \frac{I_{in}}{2} (1 + \cos(\beta \Delta L))$$

where  $\Delta L = L_1 - L_2$  is the path length difference between the two waveguide branches. As the equation indicated the output intensity would be an oscillatory function with respect to the phase term  $\beta \Delta L$ .

The wavelength spacing between two peaks of the output intensity oscillation is called free spectral range and can be defined as

$$FSR = \frac{\lambda_o^2}{\Delta L n_g}$$

where  $n_g$  is the group index.

### 3. Modelling and simulation

#### 3.3 Waveguide characteristics

In the current design, a strip waveguide with thickness 220nm and width 500nm has been chosen for single TE mode operation at free space wavelength of 1550nm. Figures 2(a) and 2(b) show the fundamental TE and TM mode supported by this waveguide geometry simulated

from Lumerical MODE. However, for the MZI characterization only TE excitation has been used for this report.

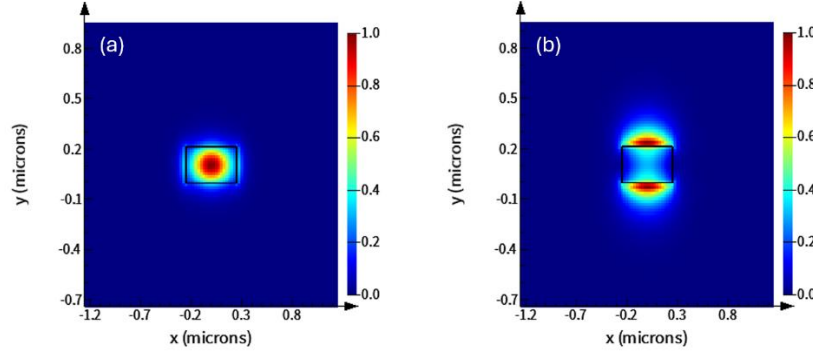


Fig 2: Electric field intensity of (a) TE mode and (b) mode for the Si waveguide of thickness 220nm and width 500nm.

Figures 3(a) and 3(b) show the effective index  $n_{eff}$  and group index  $n_g$  for the TE mode shown in Figure 2(a) a range of wavelengths from 1.5um to 1.6um. As the wavelength increases effective index decreases and group index increases for the TE mode of the waveguide under consideration.

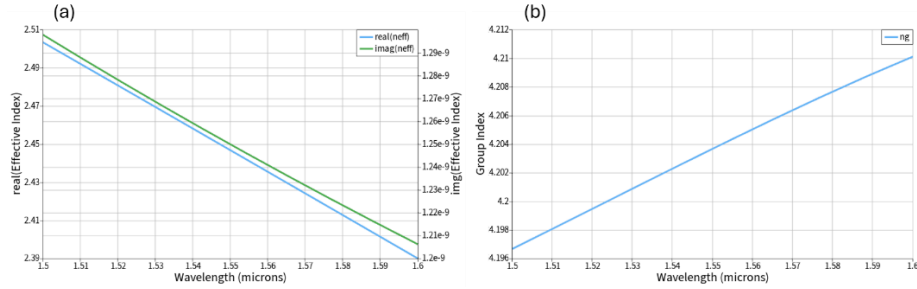


Fig 3: (a) Effective index  $n_{eff}$  and (b) group index  $n_g$  vs wavelength  $\lambda$  for the Si waveguide of thickness 220nm and width 500nm.

A compact model for the waveguide can be represented as a Taylor expansion around the center wavelength, of the form:

$$n_{eff} = n_1 + n_2(\lambda - \lambda_o) + n_3(\lambda - \lambda_o)^2$$

Where  $n_2$  and  $n_3$  has units of  $1/\mu m$  and  $1/\mu m^2$ . The polynomial coefficients have been obtained from a curve fitting of  $n_{eff}$  vs  $\lambda$  around  $\lambda_o = 1.55\mu m$  using Lumerical MODE built-in script. The model for the waveguide under consideration is stated below:

$$n_{eff} = 2.45 + (\lambda - 1.55) - 0.04(\lambda - 1.55)^2$$

### 3.2 Mach-Zehnder Interferometer (draft layout)

An initial layout draft of a few MZIs is shown in figure 4. For the splitter/combiner both the Y-splitter and broadband directional coupler have been used in different designs. Directional coupler designs have different path length difference of  $\Delta L = 25\mu m, 40\mu m$  &  $60\mu m$  and

designs with Y-splitter have different path length difference of  $\Delta L = 25\mu m, 40\mu m, 60\mu m, 100\mu m, 140\mu m$  &  $190\mu m$ . The layout also includes a design with two fiber grating couplers connected by a waveguide, to calibrate the measurement system's insertion loss.

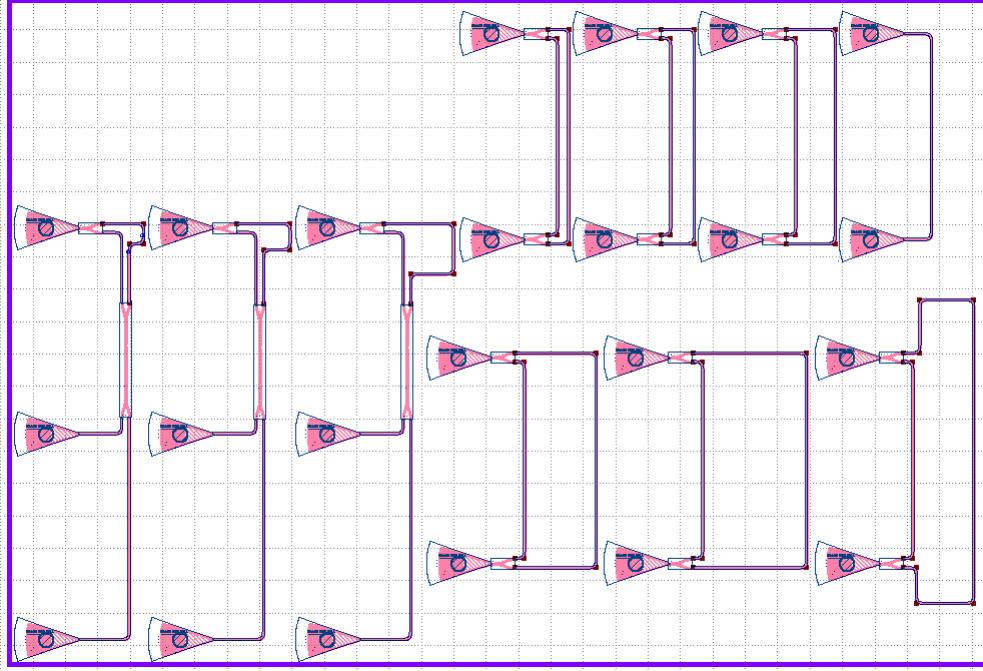


Fig 4: Initial layout draft of a few MZI designs.

### 3.3 Mach-Zehnder Interferometer characteristics

Transmission spectrum in dB measured at two output terminals for MZI based on directional coupler are shown in figures 5 (a,b,c) for three different  $\Delta L = 25\mu m, 40\mu m$  &  $60\mu m$ . The two output terminals show complementary behavior with one showing constructive interference and the other showing destructive at the same wavelength. For larger  $\Delta L$ , the wavelength gap between two consecutive peaks (namely FSR) gets reduced as well. Figure 5(d) shows the spectrum comparison between MZI based on directional coupler and Y-splitter. Compared to directional coupler, Y-Splitter only allows one output. However, Y-Splitter based MZIs also show similar trend in spectrum with smaller FSR for larger  $\Delta L$  as shown in figure 5(f)

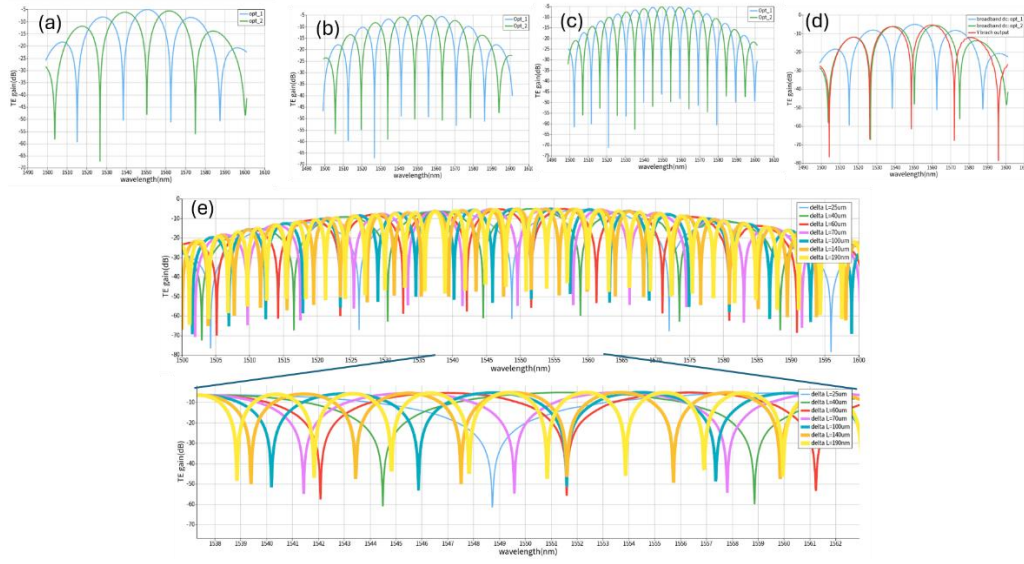


Fig 5: Transmission spectrum in dB for MZI based on directional coupler for (a)  $\Delta L = 25\mu m$  (b)  $\Delta L = 40\mu m$  and (c)  $\Delta L = 60\mu m$ . (d) Transmission spectrum comparison between MZI based on directional coupler and Y-splitter for  $\Delta L = 25\mu m$ . (e) Transmission spectrum in dB for MZI based on Y-splitter for  $\Delta L = 25\mu m, 40\mu m, 60\mu m, 100\mu m, 140\mu m$  &  $190\mu m$ .

Figure 6(a) and (b) shows the FSR obtained from Lumerical Interconnect solver for MZI based on Y-Splitter and broadband directional coupler for different  $\Delta L$ . As the formula depicts, for larger  $\Delta L$ , FSR gets smaller at a fixed wavelength. Table 1 shows the FSR value calculated from the formula (with group index  $\sim 4.2$ ) and obtained from the Interconnect solver. The two numbers match very well for both types of MZI. However, for Y-splitter FSR from simulation starts to deviate as  $\Delta L$  gets lower  $\sim 25\mu m$ .

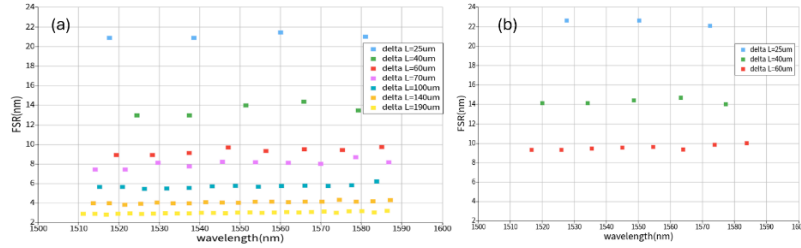


Fig 6: FSR from Lumerical simulation for (a) Y-Splitter based MZI and (b) directional coupler based MZI at different  $\Delta L$

**Table 1. FSR theoretical vs simulation**

| $\Delta L(\mu m)$ | FSR (nm)             |            | FSR (nm)               |            |
|-------------------|----------------------|------------|------------------------|------------|
|                   | Y-Splitter based MZI |            | Broadband DC based MZI |            |
|                   | Theoretical          | Simulation | Theoretical            | Simulation |
| 25                | 22.86                | 21.4       | 22.86                  | 22.8       |
| 40                | 14.28                | 14         | 14.28                  | 14.3       |
| 60                | 9.5                  | 9.5        | 9.5                    | 9.5        |
| 100               | 5.7                  | 5.8        | 5.7                    |            |

|     |   |   |   |
|-----|---|---|---|
| 140 | 4 | 4 | 4 |
| 190 | 3 | 3 |   |

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The group index can also be inferred from the FSR number obtained from simulation as  $n_g = \frac{\lambda_o^2}{FSR \Delta L}$ . The equation can be tested as using  $\lambda_o = 1.55\mu m$ ,  $\Delta L = 40 \mu m$  and  $FSR = 14$ ,  $n_g$  is found to be 4.29. Experimentally, FSR can be determined from the transmission spectrum for MZI at a known  $\Delta L$ . Thus,  $n_g$  can be calculated using the measured parameters at a specific wavelength.

#### 4. Fabrication

#### 5. Experiment data

#### 6. Conclusion

#### 7. References

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